PROBLEM 8 · WHITEBOARD GUIDE

Cooling a steel billet, by hand

A board-by-board guide for teaching 2D time marching, explicit and implicit (ADI), with a pocket calculator, before showing the code

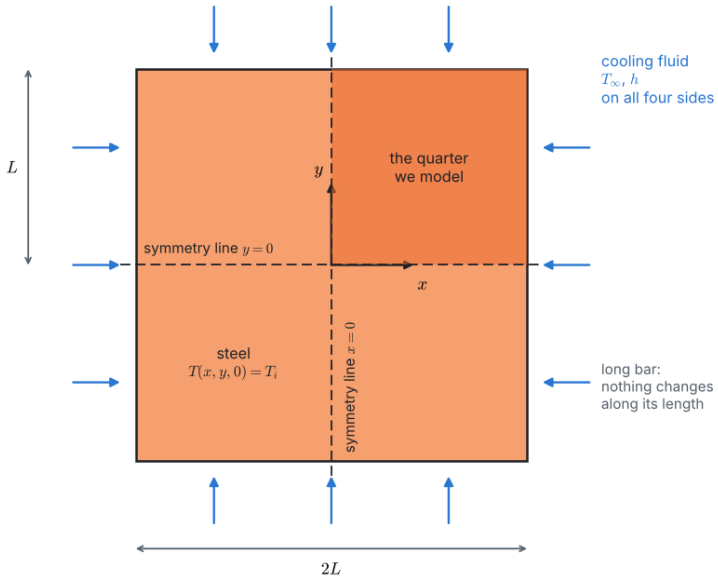
Aim of the lesson. Students see a 2D temperature field evolve in time, step by step. On the coarsest possible grid (one quarter of the billet, $N = 2$, 3 × 3 nodes) they first check that lumped capacitance fails, derive the node equations from energy balances, find the time-step limit set by the cooled corner, and march **two explicit steps** by hand. Then they take **one ADI step** five times as long, with the Thomas algorithm of Problem 1, and compare both with the exact solution. Every number below can be checked against the notebook cell under *Checking the whiteboard numbers*.

Lesson plan (about 70 minutes)

Board	Content	Time
1	The billet, the Biot number, and why one temperature is not enough	7 min
2	Symmetry, the quarter, and the 3 × 3 grid	6 min
3	Energy balances: the node equations and the time-step limit	12 min
4	FTCS step 1: only the surface feels the cooling	6 min
5	FTCS step 2: the cooling moves inwards	10 min
6	ADI, half step 1: one tridiagonal system per row	12 min
7	ADI, half step 2, and the comparison with the exact solution	12 min
8	Questions and discussion	5 min

Calculator advice for the class. Work with the **excess temperature** $\theta = T - T_\infty = T - 30$ °C: the cooling fluid is then at $\theta = 0$ and every constant T_∞ term disappears. The billet starts at $\theta = 570$. Keep **3 decimals**. Use the **symmetry** of the quarter about its diagonal: node (i, j) has the same temperature as node (j, i), so only 6 of the 9 values must be computed in each FTCS step.

Board 1 · The billet and the Biot number



The billet cross-section, cooled on all four sides.

Data: $2L = 100$ mm, $T_i = 600$ °C, $T_\infty = 30$ °C, $h = 1000$ W/(m²·K), steel $k = 48$ W/(m·K), $\rho = 7854$ kg/m³, $c_p = 559$ J/(kg·K).

$$\alpha = \frac{k}{\rho c_p} = \frac{48}{7854 \times 559} = 1.093 \times 10^{-5}\ \mathrm{m^2/s}.$$

Ask the class: *Can we treat the billet as one temperature?* Lumped capacitance needs $Bi = hL_c/k < 0.1$, with $L_c = V/A$. Per metre of billet, $V = 0.1^2 = 0.01$ m³ and $A = 4 \times 0.1 = 0.4$ m², so $L_c = 0.025$ m:

$$Bi = \frac{1000 \times 0.025}{48} = 0.52 \quad \gg \quad 0.1.$$

No. The surface will be much colder than the centre. (If we did it anyway: $\tau = \rho c_p L_c / h = 109.8$ s and $t = \tau \ln(570/70) = 230$ s to reach 100 °C. Remember this number for the end of the lesson.)

Board 2 · Symmetry and the grid

The billet is symmetric about its two centre lines: no heat crosses them. We compute **one quarter**, with the centre of the billet at the origin.

Grid: $N = 2$ intervals per side, $\Delta x = \Delta y = 25$ mm, nodes (i, j) with $i, j = 0, 1, 2$:

	$i = 0$	$i = 1$	$i = 2$
$j = 2$ (cooled surface)	surface	surface	**corner**
$j = 1$	symmetry	interior	surface
$j = 0$ (symmetry line)	**centre**	symmetry	surface

All 9 temperatures are unknown. By the diagonal symmetry there are only 6 different values: the centre $(0, 0)$, the symmetry node $(1, 0) = (0, 1)$, the interior node $(1, 1)$, the surface middle $(2, 0) = (0, 2)$, the surface node $(2, 1) = (1, 2)$ and the corner $(2, 2)$.

Board 3 · Energy balances and the time-step limit

For each control volume: *(heat stored in one time step) = (heat flowing in, at the old time level)*. With

$$r = \frac{\alpha\, \Delta t}{\Delta x^2}, \qquad Bi_{cell} = \frac{h\, \Delta x}{k} = \frac{1000 \times 0.025}{48} = 0.52083,$$

and in θ (so that T_∞ disappears):

node	control volume	FTCS update
interior $(1,1)$	full cell	$\theta + r\,(\theta_E + \theta_W + \theta_N + \theta_S - 4\theta)$
symmetry $(1,0)$	half cell, or ghost $\theta_S = \theta_N$	$\theta + r\,(\theta_E + \theta_W + 2\theta_N - 4\theta)$
centre $(0,0)$	quarter cell, two ghosts	$\theta + r\,(2\theta_E + 2\theta_N - 4\theta)$
surface $(2,1)$	half cell	$\theta + r\,(2\theta_W + \theta_N + \theta_S - 4\theta) - 2\,r\,Bi_{cell}\,\theta$
surface middle $(2,0)$	quarter cell	$\theta + r\,(2\theta_W + 2\theta_N - 4\theta) - 2\,r\,Bi_{cell}\,\theta$
corner $(2,2)$	quarter cell, two cooled faces	$\theta + 2r\,(\theta_W + \theta_S - 2\theta) - 4\,r\,Bi_{cell}\,\theta$

Derive one on the board, the corner: a quarter cell $\frac{\Delta x}{2} \times \frac{\Delta x}{2}$, heat from W and S through faces $\Delta x/2$ long, heat to the fluid through two faces $\Delta x/2$ long:

$$\rho c_p \frac{\Delta x^2}{4}\,\frac{\theta^{n+1} - \theta}{\Delta t} = k\,\frac{\Delta x}{2}\,\frac{\theta_W - \theta}{\Delta x} + k\,\frac{\Delta x}{2}\,\frac{\theta_S - \theta}{\Delta x} - 2\,h\,\frac{\Delta x}{2}\,\theta.$$

Multiply by $4\Delta t/(\rho c_p \Delta x^2)$ to get the last row of the table.

The time-step limit. The weight of θ itself must not be negative:

$$\text{interior: } 1 - 4r \geq 0, \qquad \text{corner: } 1 - 4r - 4r\,Bi_{cell} \geq 0 \implies r \leq \frac{1}{4(1 + Bi_{cell})} = \frac{1}{4 \times 1.52083} = 0.164.$$

Point out: the corner is stricter than the interior (0.164 against 0.25), because it loses heat through two faces but stores heat in only a quarter cell.

We take $r = 0.14$, so $\Delta t = 0.14 \times 0.025^2/(1.093 \times 10^{-5}) = 8.00$ s. The numbers we need:

$$4r = 0.56, \qquad 2r = 0.28, \qquad 2\,r\,Bi_{cell} = 0.145833, \qquad 4\,r\,Bi_{cell} = 0.291667.$$

Board 4 · FTCS step 1 ($t = 8$ s)

Start: $\theta = 570$ at every node. **Ask the class:** *Which nodes change in the first step?* Wherever all neighbours are equal, every bracket is zero. Only the **convection terms** act:

node	calculation	new θ	T [°C]
centre, symmetry, interior	brackets are 0, no convection	570	600
surface middle $(2,0)$, surface $(2,1)$	$570 - 0.145833 \times 570 = 570 - 83.125$	**486.875**	516.875
corner $(2,2)$	$570 - 0.291667 \times 570 = 570 - 166.25$	**403.750**	433.750

Point out: in one step only the surface feels the cooling, and the corner (two cooled faces) feels it twice as much.

Board 5 · FTCS step 2 ($t = 16$ s)

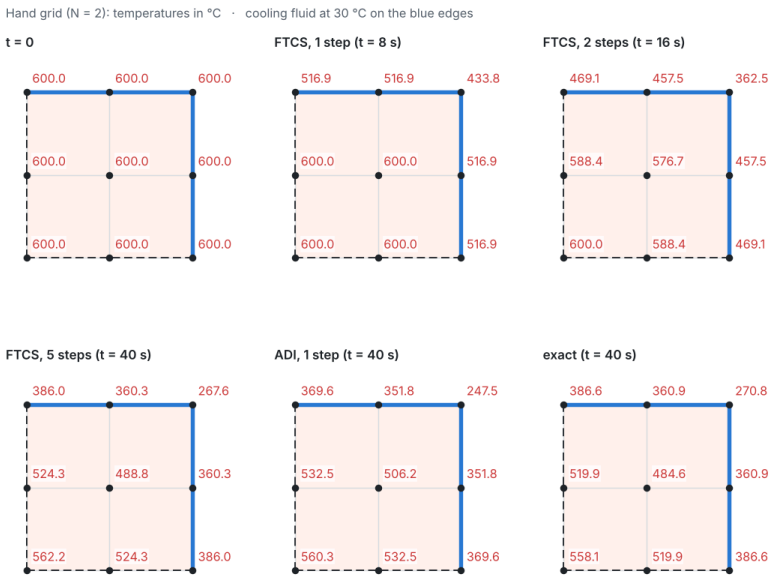
Use **only the values of step 1** on the right-hand side (explicit: write the new values in a new table).

node	calculation	new θ	T [°C]
centre $(0,0)$	$570 + 0.14\,(2\times570 + 2\times570 - 4\times570)$	570.000	600.000
symmetry $(1,0)$	$570 + 0.14\,(486.875 + 570 + 2\times570 - 2280) = 570 - 11.6375$	**558.3625**	588.3625
interior $(1,1)$	$570 + 0.14\,(486.875 + 570 + 486.875 + 570 - 2280) = 570 - 23.275$	**546.725**	576.725
surface middle $(2,0)$	$486.875 + 0.14\,(2\times570 + 2\times486.875 - 4\times486.875) - 0.145833\times486.875$	**439.147**	469.147
surface $(2,1)$	$486.875 + 0.14\,(2\times570 + 403.75 + 486.875 - 4\times486.875) - 71.003$	**427.510**	457.510
corner $(2,2)$	$403.75 + 0.28\,(486.875 + 486.875 - 2\times403.75) - 0.291667\times403.75$	**332.540**	362.540

For the surface middle: $0.14\times166.25 = 23.275$ and $0.145833\times486.875 = 71.003$, so $486.875 + 23.275 - 71.003 = 439.147$.

For the corner: $0.28\times166.25 = 46.55$ and $0.291667\times403.75 = 117.760$, so $403.75 + 46.55 - 117.760 = 332.540$.

Point out: the cooling has moved one node inwards (the symmetry and interior nodes now change); the centre still has not noticed. Heat conduction needs time: on this grid the centre changes from step 3 on.



The hand grid: two FTCS steps, then FTCS after 5 steps, one ADI step of the same 40 s, and the exact solution.

Board 6 · ADI, half step 1: implicit along the rows

Ask the class: *Can we take one step of 40 s instead of five of 8 s?* With FTCS $r = 0.70 > 0.164$: unstable. An **implicit** method can. ADI splits the step into two halves; in the first, the x direction is implicit and the y direction explicit. With $s = r/2 = 0.35$, for each row j:

$$\theta_i^* - s\,\delta_x^2\theta_i^* = \theta_i + s\,\delta_y^2\theta_i,$$

with the second differences of Board 3: $\delta^2\theta = 2(\theta_1 - \theta_0)$ at a symmetry node, $\theta_{i+1} - 2\theta_i + \theta_{i-1}$ inside, $2(\theta_1 - \theta_2) - 2Bi_{cell}\theta_2$ at a cooled node. The same 3 × 3 tridiagonal matrix serves every row:

$$\begin{bmatrix} 1.7 & -0.7 & \\ -0.35 & 1.7 & -0.35 \\ & -0.7 & 2.064583 \end{bmatrix} \begin{bmatrix} \theta_0^* \\ \theta_1^* \\ \theta_2^* \end{bmatrix} = \text{right-hand side}, \qquad 2.064583 = 1 + 0.7\,(1 + 0.52083).$$

Thomas algorithm (Problem 1). The forward-sweep factors depend only on the matrix, so compute them once:

$$c_0^* = \frac{-0.7}{1.7} = -0.411765, \qquad d_1 = 1.7 - (-0.35)(-0.411765) = 1.555882, \qquad c_1^* = \frac{-0.35}{1.555882} = -0.224953,$$

$$d_2 = 2.064583 - (-0.7)(-0.224953) = 1.907116.$$

Then for every right-hand side: $r_0^* = \mathrm{rhs}_0/1.7$, $r_1^* = (\mathrm{rhs}_1 + 0.35\, r_0^*)/1.555882$, $r_2^* = (\mathrm{rhs}_2 + 0.7\, r_1^*)/1.907116$, and back: $\theta_2 = r_2^*$, $\theta_1 = r_1^* + 0.224953\,\theta_2$, $\theta_0 = r_0^* + 0.411765\,\theta_1$.

Right-hand sides (explicit in y, from $\theta = 570$ everywhere): in rows $j = 0$ and 1, $\delta_y^2\theta = 0$, so every entry is 570. In row $j = 2$ (the cooled surface), $\delta_y^2\theta = 2(570 - 570) - 2 \times 0.52083 \times 570 = -593.75$, so every entry is $570 - 0.35 \times 593.75 = 362.1875$.

row	right-hand side	$r_0^*,\ r_1^*,\ r_2^*$	$\theta_0^*,\ \theta_1^*,\ \theta_2^*$
$j = 0$ and $j = 1$	570, 570, 570	335.294, 441.777, 461.033	**559.907, 545.488, 461.033**
$j = 2$	362.1875 (×3)	213.051, 280.712, 292.948	**355.774, 346.612, 292.948**

Point out: after this half step the cooling has reached **every node of a row at once**, even the centre (559.907 < 570). In an implicit method information crosses the whole grid in one step. The intermediate field θ^* is not symmetric (only x was treated implicitly); the second half step restores the symmetry.

Board 7 · ADI, half step 2, and the comparison

Now implicit along each **column** i, explicit in x (using θ^*): $\theta_j^{n+1} - s\,\delta_y^2\theta_j^{n+1} = \theta_j^* + s\,\delta_x^2\theta_j^*$. Same matrix, same factors.

Right-hand sides, for example column 0, row 0: $\delta_x^2\theta^* = 2(545.488 - 559.907) = -28.838$, so $559.907 - 0.35 \times 28.838 = 549.813$. Column 2, row 0 (a cooled node): $\delta_x^2\theta^* = 2(545.488 - 461.033) - 1.041667 \times 461.033 = -311.333$, so $461.033 - 108.967 = 352.066$.

column	right-hand side ($j = 0, 1, 2$)	$r_0^*,\ r_1^*,\ r_2^*$	θ^{n+1} ($j = 0, 1, 2$)
$i = 0$	549.813, 549.813, 349.361	323.420, 426.131, 339.598	**530.342, 502.525, 339.598**
$i = 1$	520.975, 520.975, 331.036	306.456, 403.780, 321.786	**502.525, 476.167, 321.786**
$i = 2$	352.066, 352.066, 223.709	207.098, 272.868, 217.457	**339.598, 321.786, 217.457**

Point out: the result **is** symmetric again: $\theta_{1,0} = \theta_{0,1} = 502.525$, $\theta_{2,0} = \theta_{0,2} = 339.598$.

Compare at $t = 40$ s (in °C, $T = \theta + 30$; the notebook did the five FTCS steps):

node	FTCS, 5 steps of 8 s	ADI, 1 step of 40 s	exact
centre $(0, 0)$	562.2	560.3	558.1
interior $(1, 1)$	488.8	506.2	484.6
surface middle $(2, 0)$	386.0	369.6	386.6
corner $(2, 2)$	267.6	247.5	270.8

> **One ADI step of 40 s (five different Thomas solves of 3 unknowns) gives the same picture as five FTCS steps**, and FTCS could not have taken that step at all. On this very coarse grid both differ from the exact solution by a few degrees inside; ADI is further off at the corner, because a single long step cannot follow the very fast initial cooling there. The notebook refines to 11 × 11 nodes, where FTCS ($\Delta t = 0.5$ s) and ADI ($\Delta t = 10$ s) both find that the centre reaches 100 °C after 349 s, as the exact solution does, and not after 230 s as lumped capacitance claimed on Board 1.

Questions for the class

- **Why does only the surface change in the first FTCS step?** (All brackets are zero in a uniform field; only the convection terms act. Information travels one node per explicit step.)
- **Why is the corner the most restrictive node?** (Two cooled faces, a quarter cell: it loses heat fastest relative to what it stores.)
- **What happens if we take $r = 0.25$ with FTCS on this grid?** (The corner weight $1 - 4r(1 + Bi_{cell}) = -0.52$ is negative. The notebook shows it oscillates and grows: unstable already at $r = 0.24$.)

- **Why is the ADI intermediate field not symmetric?** (In the first half step only x is implicit; the second half step treats y the same way and restores the symmetry.)
- **ADI is unconditionally stable. Can we take one step of 600 s?** (It stays bounded, but it is inaccurate, and the corner temperature oscillates. Stability is not accuracy.)
- **Why does lumped capacitance give a much shorter time?** (It uses the average temperature; the centre is the hottest point and the last to cool.)

Common mistakes to watch for

- Forgetting the factor **2** for the ghost nodes on the symmetry lines, or for the neighbour across from a cooled face.
- Using **new** values on the right-hand side of an FTCS step.
- Forgetting the convection term ($-2rBi_{cell}\theta$, or $-4rBi_{cell}\theta$ at the corner), or its sign.
- In ADI: putting the implicit direction on the right-hand side, or forgetting the explicit δ^2 term of the other direction.
- In the Thomas algorithm: the off-diagonals are **negative**, so $-a_i\, r^*_{i-1}$ becomes $+0.35\, r^*_0$ and $+0.7\, r^*_1$.

From the board to the notebook

`03_fdm_solver.ipynb` repeats every board: the Biot-number check of Board 1 is the lumped-capacitance cell; the table of Board 3 is `second_difference` and `ftcs_step`; the limits of Board 3 are `limits`; Boards 6 and 7 are `adi_step`, which calls the `thomas` function of Problem 1. The cell under *Checking the whiteboard numbers* prints every field of Boards 4 to 7, and the notebook then refines the grid to 11 × 11 nodes (and up to 41 × 41 for the convergence study).